SAMPLE

Solar System Missions: SOHO

Solar and Heliospheric Observatory

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Introduction

SOHO, the Solar and Heliospheric Observatory, is the spacecraft that has watched the Sun longer than any other. Launched on the second of December 1995 by a joint European-American team and parked at a gravitational balance point a million and a half kilometers toward the Sun from Earth, SOHO has stared at our star continuously across three decades and more than two full solar cycles, the longest unbroken study of any astronomical object ever made, the machine that turned the Sun from a light in the sky into a patient, monitored, and largely understood neighbor.

The observatory's own story is the trade's best yarn: a spacecraft nearly lost in 1998, its gyroscopes failing, the contact gone silent for months, and then found, rescued, and rebuilt in flight, the deepest save in the history of the business, and the mission that came back to discover thousands of comets, image the invisible corona, record the great storms of the space-weather era, and sound the interior of the star itself. The chapters ahead tell it in order: the origins and the launch, the Lagrange point and the twelve instruments, the science of the Sun that the observatory was built to do, the loss and the rescue, the comet discoveries and the space-weather service, and the long twilight of a mission designed for two years and still working in its fourth decade.

SOHO matters for the simplest of reasons: everything the Sun does, it does to us. The star's wind and its storms wash over the planet daily, the auroras and the satellite outages and the power-grid worries all downstream of the corona, and SOHO was built to watch the source, the first sentry of the space weather the modern world now plans around. This book is the account of that watching, the observatory at the Sun's elbow, the species' steady eye on its own star.

A note on the numbers: the dates and counts in this account, the launch and the loss, the comet tally and the mission's years, are the public record's own, stated as the agencies state them, and where the record's precision ends the text says so, since a mission still flying keeps its own ledger open, the counts still rising, the story still in its present tense, which is the account's own honesty, a book about a living mission, told while it lives.

The chapters ahead keep the watch's own calendar, the story told as the mission lived it: the early chapters the origins, the station, and the machine with its dozen eyes, the middle chapters the flight, the loss, and the rescue, the later chapters the findings, the comets, the storms, and the mission's long afterlife in the field and the public. The mission in a single sentence: a stubborn spacecraft has stared at the Sun without a miss for three decades, died once, come back, and along the way read the star's inside, counted its storms, found its comets, and taught the modern world to plan around its weather, the whole Sun, watched, the longest stare in history, still going, right now, the account the following chapters now tell.

Chapter 1: The Discovery of SOHO

The discovery story of SOHO begins with a gap in the watching: the Sun, the one star the species depends on utterly, was being observed in fragments, the ground telescopes blinded by the atmosphere, the earlier spacecraft brief or partial, and the questions, the corona's heat, the wind's engine, the interior's structure, all waiting for a mission that could stare without interruption.

The mission was conceived as a joint project from the start, the European Space Agency building the spacecraft and NASA launching it and tracking it, the two agencies pooling the solar science of two continents into one observatory, a division of labor the era's budgets made natural and the mission's history made famous. The spacecraft was assembled in Europe, the twelve instruments built by the teams of many nations, and the whole package, two and a half tons of solar physics, was launched on an Atlas rocket on the second of December 1995, the trade's biggest solar mission on the era's standard truck.

The destination was the mission's first innovation: not an orbit around Earth, where the planet's shadow and the telescopes' vibration would interrupt the watching, but the first Lagrange point, the gravitational balance four times farther out than the Moon, on the sunward side, where the pulls of the two bodies cancel and a spacecraft can hang in a halo orbit with the Sun always in view. SOHO was the first solar mission to use the point, arriving in the spring of 1996 and settling into the halo path the mission has held ever since, the continuous view achieved, the Sun watched without a blink, the mission's founding idea, made real.

The first light came in the new year of 1996, the instruments opening their eyes, and the discovery the mission delivered was immediate: the Sun whole, the disk and the corona and the wind's birth in one continuous gaze, the

star's weather visible for the first time as weather, the eruptions and the holes and the waves of the corona arriving in the data as the missions' teams had only imagined them. What follows is the record of what the watching found, the observatory that never blinks, and the discoveries that the patient stare has earned, the Sun at last, kept.

The discovery chapter should credit the mission's builders, the two-continent team behind the watch: the European spacecraft engineers who built the observatory, the instrument teams of many nations who built the eyes, and the operations teams on both sides of the ocean who have flown the mission across its decades, the human infrastructure of the long watch. The chapter's close belongs to them, since the discovery of SOHO is partly the discovery of what a patient team can do with a patient spacecraft, and the reader who wants the mission's human story can find it in the two agencies' annals, the joint project that worked, the partnership that has outlasted its members' careers, the watch, kept by people, for people, the whole way down.

The discovery chapter's closing note is the mission's timing in the solar cycle, the star's own calendar: SOHO launched near a minimum, the quiet Sun the first years watched, the calibration done on a docile star before the maximum arrived, the discovery's good luck, the observatory commissioned in the calm and tested by the storm. The paragraph is the chapter's final frame, the watch's beginning placed in the star's rhythm, and the reader who follows the mission's early images, the clean disk of 1996, can see the timing's gift, the Sun, on its best behavior, for the new instrument's first look, the politeness the maximum of the early 2000s would end, and the watch, ready, when it did.

The discovery chapter's spare entry is the mission's own address book, the watch's neighbors: the balance point SOHO shares with the fleet, the other sentries that have joined it sunward over the years, the gravitational parking the trade's solar missions now use, the discovery's modern echo, the point the mission pioneered now a busy place. The entry is the chapter's forward glance, the address the trade filled, and the reader who tracks the current fleet at L1 is looking at the mission's legacy in real estate, the neighborhood, founded by the watch, the discovery chapter's closing aside, the point, popular, the Sun's doorstep, crowded now, thanks to the mission that proved it, the first resident, still there, the oldest neighbor, at the point, waving, quietly, in telemetry.

The fleet context belongs in the origins story too, because SOHO was launched as one of a set: the International Solar-Terrestrial Physics program, the era's coordinated assault on the Sun-Earth connection, sent SOHO to watch the source while sibling spacecraft monitored the wind and the Earth's own magnetosphere, the observatory the flagship of a formation. The reader who weighs the mission should weigh its design position, not a lone watcher but the flagship of a convoy, its data meant to be read alongside the others' from the first day, the origin's own signature, a team player in the largest sense, the watch, from birth, part of a bigger watch.

The era's questions deserve their own paragraph before the mission is described, because SOHO was built against a list the field had carried for decades: why the corona is hundreds of times hotter than the surface beneath it, what accelerates the solar wind on its way out, and what the Sun's interior actually looks like beneath the light we can see. The three briefs set the instrument manifest, the imagers and the listeners and the particle sensors all bought against a named ignorance, and the reader who wants the mission's logic should start with the list, the spacecraft as an answer with a question attached, the trade's honest way of building, the ignorance, priced, the observatory, the purchase.